Agustín Nolasco

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About Me

I'm a Computer Science graduate student at the National University of Río Cuarto (UNRC), Argentina. My research interests include software testing, formal methods, and automated verification. I have experience in academic research projects and a strong foundation in programming and software development. Currently, I am looking for opportunities to apply my skills and knowledge in real-world scenarios, particularly in the areas of software quality assurance and verification. Anyway, I can really adapt to different environments and challenges, so feel free to reach out if you think I could be a good fit for your team or project.

Employment History

Ago. 2024 – Present
Teaching Assistant, National University of Río Cuarto, Argentina
Courses: Compilers – Design of Algorithms – Algorithms and Data Structures

Apr. 2024 – Aug. 2024
Student Teaching Assistant, National University of Río Cuarto, Argentina
Course: Advanced Programming

Mar. 2024 – Aug. 2024
QA Tester, Oliver Informatics
Río Cuarto, Córdoba, Argentina

Apr. 2023 – Jan. 2024
Grader, Guangdong Technion – Israel Institute of Technology, China
Course: Introduction to Computer Science

Apr. 2023 – Jul. 2023
Grader, Guangdong Technion – Israel Institute of Technology, China
Course: Introduction to System Programming

Apr. 2022 – Sept. 2022
Student Teaching Assistant, National University of Río Cuarto, Argentina
Course: Design and Analysis of Algorithms

Education

2024–Present
PhD. in Computer Science in National University of Córdoba, FAMAF. Córdoba.
Ongoing.

2019–2024
Computer Science Licentiate in National University of Río Cuarto. Córdoba.
Final Average Score: 9.88 out of 10.
Thesis title: *Inference of Equivalences of Method Sequences for Metamorphic Testing*.

22019–2021
Computing Analyst in National University of Río Cuarto. Córdoba.
Final Average Score: 9.81 out of 10.

2012–2018
Electromechanical Equipments and Installations Technician in Technical School No.15 Ing. Agustín Mercau. Villa Mercedes, San Luis.
Final Average Score: 8.65 out of 10.

Research Publications

Journal Articles

1 A. Nolasco, F. Molina, R. Degiovanni, *et al.*, "Abstraction-aware inference of metamorphic relations," *Proc. ACM Softw. Eng.*, vol. 1, no. FSE, Jul. 2024. DOI: 10.1145/3643747.

Preprints

1. R. Degiovanni, F. Molina, A. Nolasco, N. Aguirre, and M. Papadakis, *Specification inference for evolving systems*, Pre-print, 2023. arXiv: `2301.12403 [cs.SE]`.

Research Projects

Dec. 2021 – Present

Optimizations to Symbolic Execution and Applications
PICT 2019
Director: Dr. Nazareno Aguirre
Position: Collaborator
Funded by: National Agency for the Promotion of Research, Technological Development, and Innovation, through FONCyT.

Jan. 2020 – Present

Application of Constraint Solving Techniques to Software Validation and Verification
PPI 2020-2022
Director: Dr. Nazareno Aguirre
Position: Collaborator
Funded by: Secretariat of Science and Technology, National University of Río Cuarto.

Skills

Languages	English: Lower Intermediate (strong reading/listening skills; basic conversational ability).
Coding	Java, C, C++, Python, bash, SQL, LaTeX, …
Databases	MySQL, PostgreSQL, SQLite.
Misc.	Academic research, teaching, Programming, UNIX, LaTeX typesetting and publishing.

Courses

Mar. 2025

Advances in Automatic Software Testing
dictated by Dr. Mike Papadakis, Río 2025 Summer School
Final Score: 9 out of 10

Introduction to Input Grammar Fuzzing
dictated by Dr. Andreas Zeller, Río 2025 Summer School
Final Score: 10 out of 10

Analysis of mobile apps for quality and trustworthiness
dictated by Dr. Alessandra Gorla, Río 2025 Summer School
Final Score: 10 out of 10

Feb. 2024

Dynamic Modal Logics
dictated by Dr. Raul Fervari and Dr. Carlos Areces, Río 2024 Summer School
Final Score: 6 out of 10

Verification of Fault Trees
dictated by Dr. Joost-Pieter Katoen, Río 2024 Summer School

Aug. 2023

Advanced Data Structures
dictated by Dr. Conrado Martínez, ECI 2023
Final Score: 10 out of 10

Fuzzing for Compiler Testing
dictated by Dr. Guillermo Polito, ECI 2023
Final Score: 9 out of 10

Courses (continued)

Feb. 2023 — **Modeling and Analyzing Concurrent Systems**
dictated by Dr. Thomas Noll, Río 2023 Summer School
Final Score: 8 out of 10

Formal Approaches to Decision-Making under Uncertainty
dictated by Dr. Arnd Hartmanns, Río 2023 Summer School

Aug. 2022 — **Introduction to APIs**
dictated by Dr. Guido De Caso, ECI 2022
Final Score: 8 out of 10

Satellite Images Processing
dictated by Dr. Rafael Grimson, ECI 2022
Final Score: 7 out of 10

Oct. 2019 — **Software Testing**
dictated by Dr. Nazareno Aguirre, CACIC 2019
Final Score: 10 out of 10

2011–2016 — **English Academy**
Superior Institute of English Language, Villa Mercedes, San Luis

Miscellaneous Experience

Awards and Achievements

2023 – 2024 — **First Major National Flag Escort**, National University of Río Cuarto.

2017 – 2018 — **National Flag Bearer**, Technical School No.15 Ing. Agustín Mercau.

2017 — **Max Average Score of the Class**, Technical School No.15 Ing. Agustín Mercau.
Average Score 9.66 out of 10

Scholarships

Ago. 2024 – Present — **PhD. Scholarship**. National Scientific and Technical Research Council (CONICET).

Sept. 2022 – Aug. 2023 — **EVC-CIN Scholarship 2021**. Stimulation Scholarship for Scientific Vocations.

References

Dr. Nazareno Aguirre
Full Professor – Independent Researcher
National University of Río Cuarto, Córdoba, Argentina
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Dr. Facundo Molina
Post-Doctoral Researcher
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